

BROOKHAVEN CALABRO, NY, USA

WMO: 725016

Lat: 40.822N

Lon: 72.869W

Elev: 25

StdP: 101.03

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54790

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.2	(c) -9.6	(d) -20.8	(e) 0.6	(f) -9.6	(g) -18.5	(h) 0.7	(i) -7.7	(j) 11.8	(k) -0.2	(l) 10.8	(m) -1.6	(n) 3.5	(o) 330	(p) 0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	31.3	23.0	29.6	22.3	28.2	21.9	25.1	28.3	24.3	27.4	23.6	26.5	4.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.0	26.6	23.3	18.1	25.9	22.7	17.4	25.4	76.3	28.1	73.4	27.5	70.5	26.5	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.7	7.8	-16.0	34.6	2.8	1.8	-18.0	35.9	-19.7	37.0	-21.3	38.0	-23.3	39.4
			-16.7	26.2	2.7	1.0	-18.6	27.0	-20.2	27.6	-21.7	28.2	-23.7	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-0.4	0.3	4.0	9.6	14.8	19.9	23.6	22.8	19.1	13.2	7.6	3.1		
	DBStd	9.26	5.36	4.73	4.57	3.95	3.70	3.44	2.64	2.77	3.34	4.35	4.39	4.74		
	HDD10.0	1181	322	273	192	54	4	0	0	0	0	18	97	221		
	HDD18.3	2934	579	506	445	264	121	23	1	2	31	168	322	473		
	CDD10.0	1738	1	1	6	41	152	297	421	398	274	117	26	6		
	CDD18.3	451	0	0	0	1	12	69	163	141	55	9	0	0		
	CDH23.3	2999	0	0	0	16	67	419	1212	1012	250	23	0	0		
	CDH26.7	732	0	0	0	4	11	106	341	237	32	3	0	0		
(14) Wind	WSAvg	3.4	3.8	3.9	4.1	3.8	3.2	3.1	2.9	2.6	2.7	3.2	3.4	3.6		
(15) Precipitation	PrecAvg	1139	90	80	113	99	87	92	83	97	104	104	83	116		
	PrecMax	1486	190	134	300	179	177	215	145	229	192	451	138	209		
	PrecMin	854	46	29	18	42	15	5	27	20	31	11	21	41		
	PrecStd	153	34	32	62	41	38	57	32	56	42	85	32	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	14.2	19.0	26.0	28.1	32.3	34.2	32.5	29.6	26.3	19.4	16.4		
		MCWB	12.0	11.0	12.6	15.5	20.1	23.3	24.2	23.6	23.4	20.1	15.7	13.7		
	2%	DB	11.6	11.5	14.9	20.9	25.1	29.4	32.0	30.7	27.4	23.1	17.7	14.2		
		MCWB	10.3	8.9	9.1	13.2	18.0	21.5	23.2	23.0	20.9	20.0	14.9	12.7		
	5%	DB	9.7	9.5	12.5	18.1	23.0	27.5	30.0	29.1	26.0	21.5	16.4	12.6		
		MCWB	7.8	7.2	9.0	11.7	17.0	20.7	22.7	22.4	20.5	18.0	13.8	11.1		
	10%	DB	7.8	7.6	10.7	15.9	21.1	25.7	28.4	27.8	24.7	20.0	14.9	11.0		
		MCWB	5.5	5.2	7.7	10.6	16.0	19.8	22.2	22.1	20.1	17.2	12.6	9.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	12.1	14.2	17.8	21.8	24.4	25.9	26.2	24.4	22.1	17.5	15.0		
		MCDB	13.2	12.9	18.0	23.2	26.7	29.5	30.9	29.7	27.6	24.2	18.2	15.8		
	2%	WB	10.5	9.8	11.3	15.1	19.9	23.1	25.0	25.3	23.3	20.4	15.9	13.1		
		MCDB	11.4	11.0	13.0	18.4	23.3	27.9	28.8	27.8	25.6	22.6	17.2	13.9		
	5%	WB	8.1	7.7	9.7	13.4	18.6	22.1	24.4	24.5	22.4	19.2	14.5	11.5		
		MCDB	9.3	9.1	11.9	16.3	21.5	25.5	27.6	27.1	24.4	20.9	16.0	12.5		
	10%	WB	6.0	5.5	8.0	12.0	17.3	21.2	23.7	23.7	21.6	17.7	13.0	9.3		
		MCDB	7.4	7.3	10.1	14.3	20.0	24.0	26.7	26.3	23.5	19.5	14.7	10.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.6	8.6	9.4	9.1	8.7	8.4	8.7	9.1	9.2	9.0	8.0		
		MCDBR	9.8	10.2	10.8	12.8	11.6	10.6	10.3	9.7	9.6	9.3	9.0	8.5		
	5% WB	MCWBR	9.1	8.5	7.6	6.8	5.9	5.0	4.0	4.2	5.3	6.1	7.3	8.6		
		MCWBR	9.4	10.0	9.6	9.7	9.5	8.8	7.9	7.0	7.0	7.5	7.9	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.317	0.337	0.383	0.434	0.476	0.490	0.466	0.396	0.348	0.318	0.309		
		taud	2.517	2.494	2.479	2.369	2.262	2.186	2.155	2.217	2.383	2.531	2.562	2.560		
	Ebn at Noon		855	901	917	889	847	809	793	801	841	854	839	828		
		Edh at Noon	74	87	99	118	134	145	148	136	108	83	70	66		
(44) All-Sky Solar Radiation	RadAvg		1.83	2.72	3.77	4.87	5.44	5.94	6.01	5.40	4.40	3.01	2.08	1.56		
	RadStd		0.15	0.26	0.32	0.50	0.53	0.57	0.31	0.34	0.35	0.28	0.17	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	+61

Nomenclature: See separate page